

Repair What Matters: Criticality-Bounded Revision Cones for Persistent Captured Worlds

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Persistent captured worlds—AR maps, digital twins, and long-lived reconstructions—continue changing after capture. Their geometry, appearance, GPU state, and temporal history become interdependent, so a small physical edit can trigger broad recomputation. We present *Criticality-Bounded Revision Cones* (CBRC), a fail-closed revision rule that combines exact structural closure, conservative finite-change bounds, output tolerances, work-aware selection, automatic FULL fallback, and independent full-after verification. We evaluate CBRC in MAVEB, a research prototype built on AETHER, a C++23 persistent-world stack for reconstruction, Gaussian rendering, revision tracking, GPU publication, temporal state, and evidence capture. A local repair is accepted only when the unrecomputed exterior stays within tolerance and the repair remains cheaper than rebuilding.

The base evaluation contains 1,275 revisions over 85 scenes spanning 3RScan, ARKitScenes, Bonn RGB-D Dynamic, and trained GraphDECO 3DGS. It yields 935 local repairs, 340 full rebuilds, zero observed certificate violations, and median calibrated selected work of 0.474 of a rebuild. We then test measurable approximation separately. An immutable diagnostic shows that the original image certificate rejects every non-zero residual. An opacity-delta Lipschitz refinement restores certified non-zero locality in a four-scene mechanism study. A separate outcome-blind confirmation repeats the unchanged mechanism on 20 independently selected scenes and 3,840 pooled parameter cases. Non-zero opacity locality and tolerance crossover occur in all 20 scenes and all four dataset families, while the translation control remains at zero. The scene-level opacity-minus-translation difference is 0.477 with a dataset-stratified bootstrap 95% interval of [0.432, 0.517], with no observed certificate violation. The certificate remains conservative, leaving certificate tightening as an explicit limitation. The result is selective maintenance tied to an output-error contract and reproducible fallback rather than heuristic locality.

CCS Concepts: • **Computing methodologies** → **Computer graphics**; **Rendering**; *Reconstruction*.

Additional Key Words and Phrases: persistent captured worlds, incremental reconstruction, 3D Gaussian Splatting, certified recomputation, change-aware rendering, dependency analysis

1 INTRODUCTION

Many 3D systems now live well beyond the capture session that created them. A reconstructed room may be revisited after furniture has moved; an AR map continues to exist while objects appear and disappear; a digital twin is updated after a small physical intervention. The stored state in these systems contains heterogeneous representations. Observations feed geometry, geometry determines surface or Gaussian support, materials and textures follow ownership and visibility, GPU records mirror persistent entities, and temporal filters reuse earlier screen-space state. For that reason, an edit that

looks local in Euclidean space can become nonlocal once it propagates through the system.

The safest response is still a global rebuild. Classical volumetric fusion and dense reconstruction give a natural full-reference notion of recomputation [7, 8, 19]. That choice is straightforward to reason about, but it becomes wasteful when only a small dependency closure is actually affected. Explicit scene representations make this mismatch particularly visible: large parts of a Gaussian map, or of a neural/hybrid reconstruction, may have no reason to change after a local revision [13, 14, 17]. Continual Gaussian systems already exploit this fact by focusing optimization on changed regions or by managing stale observations over time [2, 29, 30]. Our concern is slightly different. Those systems mainly decide *where to update*; here we ask how much error may safely remain outside the region that was chosen, across several kinds of derived state at once.

The question is deliberately narrow: *when is it defensible to repair less than the whole captured world while keeping an explicit output-error contract?* Reachability alone is insufficient. Ownership, record identity, publication, and unstable temporal invalidation are structural relations that require exact treatment. Other relations, such as edited Gaussian support or stable temporal blending, have finite influence that can be bounded. The revision policy must distinguish the two and fall back to a full rebuild when the remainder outside a proposed repair region cannot be bounded tightly enough.

Our answer is *Criticality-Bounded Revision Cones* (CBRC). A revision is represented on a typed dependency graph with three edge meanings. HARD edges carry exact structural requirements; ANALYTIC edges carry conservative finite-change bounds; and EMPIRICAL edges may help decide what to try first, but they never become part of the safety certificate. For a proposed cone C , we certify the unrepaired exterior O against the declared quantities of interest. The worst normalized certificate-to-tolerance ratio is the *revision criticality*, with one as the decision boundary. We accept a local result only when exact predecessors are closed, every protected output is at or below that boundary, and the calibrated regional work is below the independently measured full baseline. If the coupling is unsupported, the residual is supercritical, the analytic assumptions are unstable, or locality simply costs too much, the planner returns FULL.

MAVEB is the research system used to evaluate CBRC; AETHER is the underlying C++23 persistent-world engine stack. AETHER owns reconstruction and Gaussian state, revision transactions, GPU publication, temporal state, and provenance/evidence capture, while a dense Python reference

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certifier and an independent full-after Gaussian oracle provide separate validation paths. Figure 1 summarizes the decision path, while Figure 2 shows the support-selection idea visually. The implementation records immutable before/after revision sidecars, deterministic certificate JSON, per-domain work, and oracle replay. Planner output alone is insufficient evidence of safety. Instead, the unrepaired exterior implied by the chosen support is compared with an independently rendered FULL-after reference and checked under the contract

$$E_q^{\text{full-ref}} \leq B_q(C) \leq \varepsilon_q, \quad (1)$$

where $E_q^{\text{full-ref}}$ is the measured residual attributable to the unrepaired exterior under the selected support, evaluated against an independent full-after rendering; $B_q(C)$ is the emitted conservative bound, and ε_q is the declared tolerance for quantity of interest q .

Before examining aggregate outcomes, we fixed the evaluation protocol and claim boundaries. Across 1,275 revisions over 85 scenes, CBRC selects 935 local repairs and 340 rebuilds with zero observed certificate violations. The study covers 3RScan, ARKitScenes, Bonn dynamic RGB-D sequences, and 13 pinned trained 3DGS scenes, with translation, rotation, uniform-scale, and opacity edits. Median calibrated selected work is 0.47447 of FULL (bootstrap 95% CI 0.46657–0.47885). Mild coupling usually preserves locality; stronger interactions increasingly trigger the fail-closed path. Every selected render matches the independent FULL-after reference on the evaluated views.

That broad campaign deliberately uses exact selected-to-full agreement, so it cannot by itself show that a bounded non-zero residual can remain local. We study that question in a second, frozen line of experiments. The first diagnostic exposes a limitation in the original image certificate; a mechanism study changes only that certificate for opacity residuals and keeps translation as a control; a final confirmatory study repeats the unchanged mechanism on five independently ranked scenes from each of the same four dataset families. This staged design keeps diagnosis, mechanism change, and confirmation separate rather than tuning one campaign until it succeeds.

Operationally, this gives long-lived AR maps, digital twins, and reconstructed environments a middle path between two costly extremes: rebuilding everything after each change or trusting a nearby update without knowing what stale state remains. The system can stop propagation when the output effect is bounded, and it can abandon locality when that justification disappears.

Our contribution is the systems rule that joins existing dependency and error-estimation ideas into one captured-world revision policy: exact structural closure, bounded exterior influence, output tolerances, work-aware selection, explicit fallback, immutable records, and independent falsification all participate in the same decision.

The contributions are:

- a fail-closed formulation for revising heterogeneous persistent captured worlds, combining exact dependency closure with conservative finite-change propagation;
- output-specific certification for selective repair, including implementation-backed Gaussian display-space and stable temporal bounds, while unsupported dependencies stay exact;
- a native work-aware planner with automatic FULL fallback, deterministic record generation, a dense reference implementation, and an independent full-after falsification path; and
- a staged evaluation that separates broad system behavior from residual-sensitive mechanism evidence: an immutable negative diagnostic, a controlled certificate change with an unchanged translation comparator, and a 20-scene confirmatory replication with scene-clustered statistics.

2 RELATED WORK

Persistent reconstruction and map maintenance. Volumetric integration gave reconstruction systems a durable way to accumulate range observations [7], and surfels offered an explicit point-oriented alternative [22]. KinectFusion later showed real-time dense fusion and tracking [19]; ElasticFusion maintained and corrected dense surfel maps without a conventional pose graph [25]; and BundleFusion reintegrated surfaces online while maintaining global consistency [8]. All of these systems make map state persistent enough that revision becomes meaningful, although their update rules are closely tied to the reconstruction or consistency mechanism itself. CBRC looks at the revision one level above that: which derived state really has to be recomputed, what may remain outside it, and at what point should the incremental attempt be abandoned.

Neural and Gaussian mapping. NeRF established continuous neural radiance fields for novel-view synthesis [18], while iMAP and NICE-SLAM brought learned implicit representations into online mapping and SLAM [23, 31]. 3D Gaussian Splatting (3DGS) moved back toward an explicit anisotropic representation while retaining real-time rendering [14]; SplatTAM and Gaussian Splatting SLAM then adapted Gaussian state to online RGB-D and monocular mapping [13, 17]. Dynamic 3D Gaussians and 4D-GS extend the representation over time [16, 27]. These representations make revision especially tangible because individual persistent primitives and their support are exposed. Still, the CBRC formulation itself is representation-agnostic.

Continual and evolving Gaussian scenes. CL-Splats separates changed and static scene components so that Gaussian optimization can stay focused while older scene states are preserved [2]. GaussianUpdate uses change-specific update strategies and visibility-aware replay for continual radiance fields [30]. GaME addresses long-term evolving RGB-D

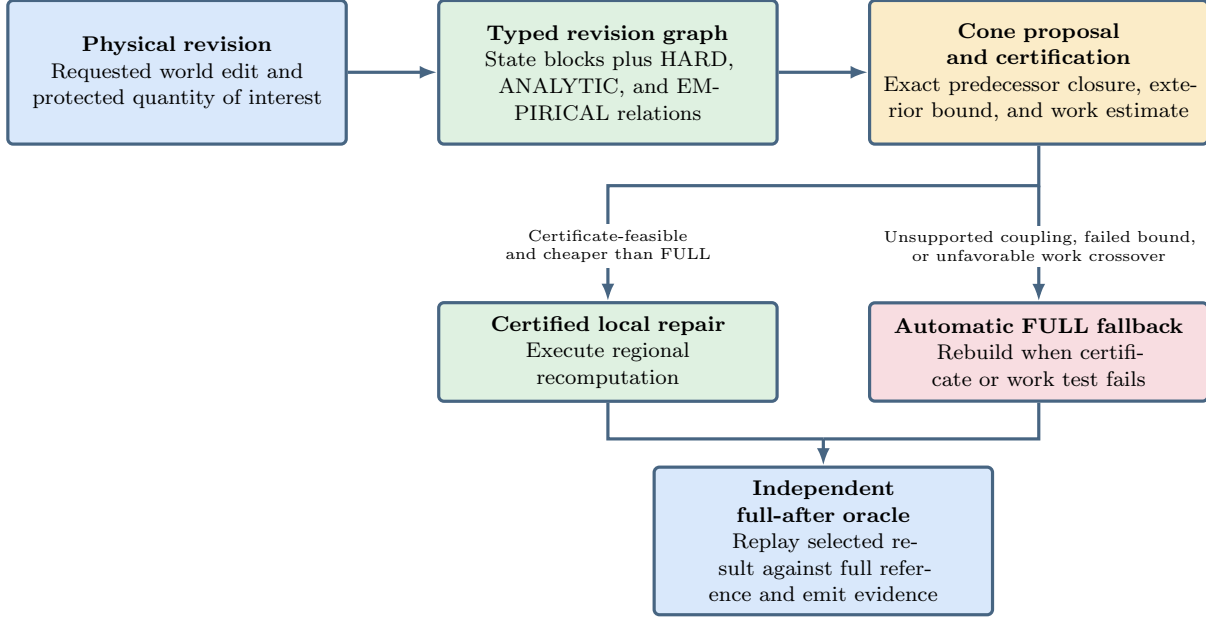


Fig. 1. CBRC decision and falsification path. A physical revision enters a typed dependency graph. Exact HARD predecessors are closed, conservative ANALYTIC influence is propagated to the unrepaired exterior, and quantity-of-interest bounds are checked against tolerance. A local repair is executed only when it is certified and cheaper than the full baseline; otherwise the system falls back to FULL. The residual outside the selected repair support is then falsified against an independently rendered full-after reference.

scenes by adapting the Gaussian map while managing stale observations [29], and EliGSiR studies continual Gaussian mapping under a bounded compute budget through adaptive view scheduling, supervision fidelity, and geometry growth [9]. We therefore do not claim the broad idea that changing Gaussian scenes should be updated locally. The distinction in CBRC is the contract: cross-representation exact dependencies and conservative output bounds are checked together, locality is admitted only under Eq. 1, and FULL remains an explicit outcome when either certification or the work comparison fails.

Editing, change detection, and identity. GaussianEditor and SC-GS demonstrate precise, controllable local editing of Gaussian representations [6, 12]. Primitive-space change detection further suggests that Gaussian attributes themselves can reveal structural and appearance changes [10], while Consistent Instance Field studies persistent probabilistic identity in dynamic Gaussian scenes [28]. These are close enough to our setting that the boundary needs to be explicit: we are not claiming local editing, primitive-level change detection, or persistent identity as new. CBRC starts after a revision has been specified or detected and asks what can safely remain unrepaired for a particular output.

Incremental rendering, change propagation, and goal-oriented error control. Dependency-aware recomputation is not new to graphics. Bala et al. connect edited world

regions to bounded-error radiance interpolants with ray segment trees [3], and Wörster et al. derive a dependency graph from scene-graph changes to rendering caches so those caches can be updated incrementally [26]. Beyond graphics, self-adjusting computation formalizes correct change propagation under mutation and reuse [1]; goal-oriented error estimation directs numerical work toward selected quantities of interest rather than a global state norm [20]. Several ingredients of CBRC clearly live in these traditions, so we treat them as novelty boundaries rather than background to be glossed over.

The mathematics used by CBRC is mostly classical. Graph closure, non-negative influence propagation, Neumann-series resolvents, norm bounds, and greedy search are not contributions of this paper. What we claim instead is a systems contract for captured worlds that brings these pieces together around one revision decision: exact dependencies across derived state, conservative finite-change bounds, rendered-output quantities of interest, a heterogeneous work crossover, fail-closed FULL fallback, and independent full-reference falsification. The mathematics is the machinery that makes that contract executable, not a standalone novelty claim.

3 PROBLEM FORMULATION

3.1 Persistent captured-world state

Let the current system state be partitioned into blocks

$$V = \{1, \dots, n\}. \quad (2)$$

How Local Gaussian Revisions Propagate

From intended physical edit to certified local repair or FULL fallback.

MAVEB / CBRC
Certified Local Revision of
Persistent Gaussian Worlds

Edit locally.
Prove the residual.
Fall back to FULL
when necessary.

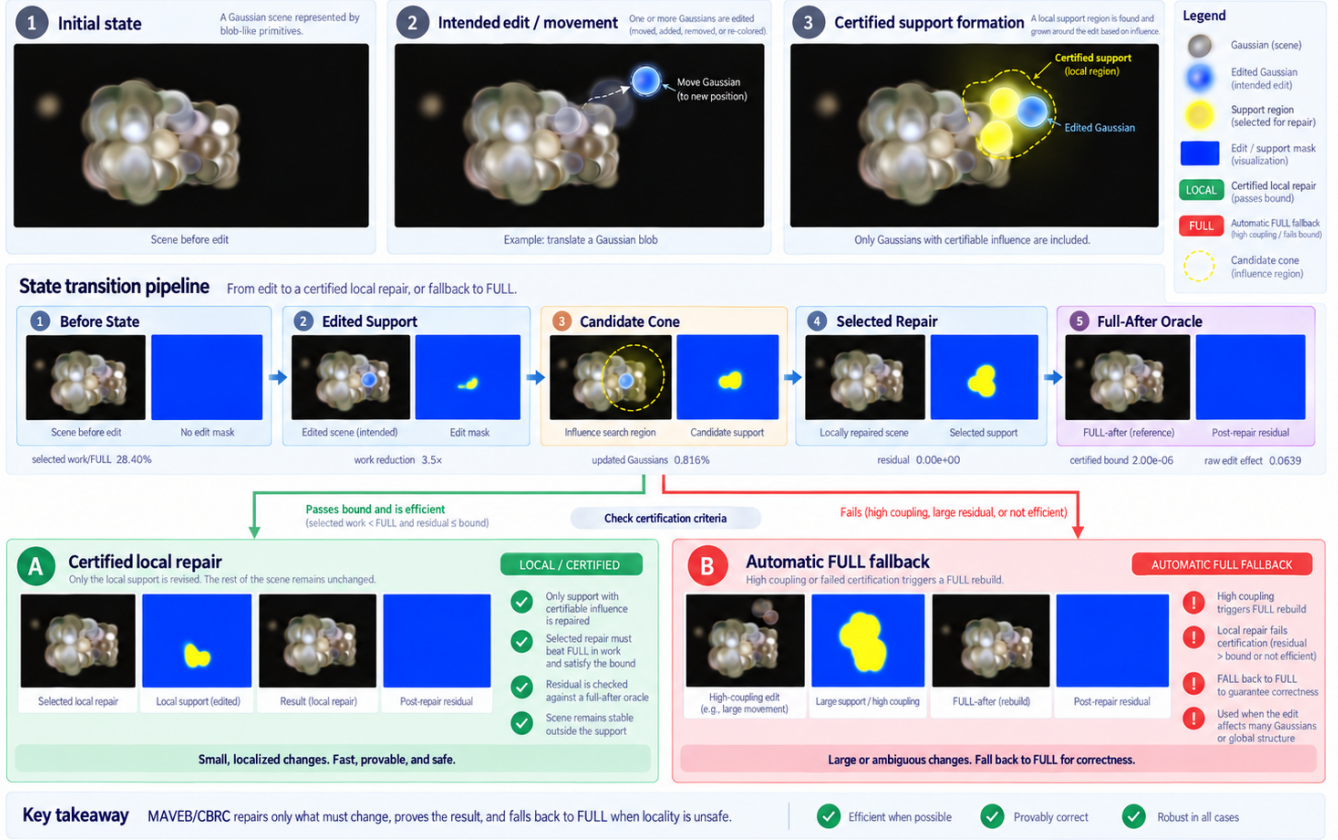


Fig. 2. How a physical edit becomes a certified revision. An intended Gaussian edit induces a candidate influence region from which CBRC selects the support that must be repaired. Exact structural dependencies remain closed and the unrepaired exterior is accepted only when its quantity-of-interest bound satisfies the declared tolerance and regional work remains below the frozen FULL baseline. The system therefore either revises only the certified support (A) or fails closed to a FULL rebuild (B). Blue marks the intended edit, yellow the selected repair support. The supplied image is reproduced unchanged. Its 28.40% work ratio, 3.5 $\times$ work reduction, and other embedded micro-metrics are illustrative rather than aggregate campaign results. The paper-grade headline is a median work ratio of 0.47447 of FULL (52.55% lower calibrated work; Section 7). Statements of correctness in the image apply only under the bound assumptions and declared revision domain; neither these work factors nor the image establish wall-clock speedup or universal safety.

A block can represent an observation region, TSDF block, mesh patch, texture page, material state, Gaussian subset, GPU publication range, temporal-history region, or another unit of derived state. It is a unit of computational ownership, not necessarily a regular spatial cell.

A physical revision gives a direct change envelope $b \in \mathbb{R}_{\geq 0}^n$. From there, the planner chooses a repair cone $C \subseteq V$ and leaves the exterior

$$O = V \setminus C \quad (3)$$

untouched. Small cones are useful, but minimizing $|C|$ by itself is not the objective. The cone has to remain feasible under the output certificate, and its measured or calibrated work must still beat the corresponding full rebuild.

3.2 Typed dependencies

Each dependency has one of three semantics (Table 1).

We keep this separation conservative on purpose. Dependencies from observations to TSDF state, TSDF to mesh ownership, meshes to texture-page identity, and Gaussian records to GPU publication remain exact in v1. Unstable temporal validation is treated the same way. By contrast, Gaussian-to-image influence and stable temporal blending have implemented ANALYTIC bounds. An EMPIRICAL-only dependency without a conservative analytic bound is promoted to exact predecessor closure for certification; its measured gain may still be used for scheduling, but never as a soft safety bound. If we do not have a conservative argument for a relation, we

Table 1. Revision-edge semantics in CBRC.

Type	Meaning	Certificate role
HARD	Exact ownership, identity, topology, publication, or invalidation dependency	Must be closed exactly
ANALYTIC	Conservative implementation-backed finite-change upper bound	May certify exterior influence
EMPIRICAL	Measured or learned influence used for ordering or scheduling	Never treated as proof

do not soften it simply because doing so would produce a smaller cone.

3.3 Output and work contracts

For each quantity of interest q , let $\varepsilon_q \geq 0$ denote its declared tolerance. A local repair is admissible only when $B_q(C) \leq \varepsilon_q$ for every protected output. That is the planner-side requirement. The experiment then asks the harder question by checking the independent full-reference contract in Eq. 1.

Locality also has to be worth doing. The planner therefore requires regional work to remain below FULL. Because the native counters describe different kinds of work, adding them directly would be meaningless. If $n_d(C)$ is native work in domain d and κ_d is the frozen calibrated cost of one native unit, the public campaign uses

$$W(C) = \sum_d \kappa_d n_d(C), \quad (4)$$

with W_{full} measured independently. The calibration is frozen before the final campaign. We keep this distinction explicit throughout the paper: Eq. 4 models heterogeneous work; it is not an end-to-end timing measurement.

4 CRITICALITY-BOUNDED REVISION CONES

4.1 Exact predecessor closure

If a repaired state block v requires an exact predecessor u , then the predecessor must also be inside the cone:

$$v \in C, u \in \text{Pred}_{\text{HARD}}(v) \implies u \in C. \quad (5)$$

This condition is checked before any approximate reasoning. If Eq. 5 is violated, the candidate is invalid regardless of how small a local sensitivity estimate happens to be.

4.2 Conservative exterior propagation

For ANALYTIC dependencies we use a componentwise non-negative influence matrix $K \in \mathbb{R}_{\geq 0}^{n \times n}$. Each entry K_{vu} upper-bounds the normalized finite change that block u can transmit to block v . Let z_C collect known change bounds for the repaired state. The unrepaired exterior then satisfies

$$\delta_O \leq b_O + K_{OC} z_C + K_{OO} \delta_O. \quad (6)$$

When the exterior response is certifiable and $\rho(K_{OO}) < 1$, the classical resolvent gives

$$G_O = (I - K_{OO})^{-1} = I + K_{OO} + K_{OO}^2 + \dots, \quad (7)$$

and therefore

$$\hat{\delta}_O = G_O (b_O + K_{OC} z_C), \quad \delta_O \leq \hat{\delta}_O. \quad (8)$$

The native v1 planner does not evaluate a general cyclic resolvent. It uses sparse finite propagation on supported DAG exteriors. If an analytic cycle falls outside that supported case, the planner fails closed: either the cone expands until the unsupported exterior disappears, or the result becomes FULL. Equation 8 should therefore be read as the reference formulation, not as a claim that arbitrary cyclic systems are handled in production.

4.3 Quantity-of-interest certificate

Let R_q map state perturbation to the protected output quantity. The exterior contribution is bounded by

$$B_q(C) = \left\| |R_{q,O}| \hat{\delta}_O \right\|_{\infty}. \quad (9)$$

A candidate is certificate-feasible only when $B_q(C) \leq \varepsilon_q$ for every declared output. As a result, the same geometric change can lead to different revision decisions when the protected outputs differ. This is intentional: the certificate is tied to what the user actually asks us to preserve, rather than to an internal state norm that may or may not correlate with visible error.

4.4 Revision criticality

We use *revision criticality* for the worst normalized certified residual of a candidate cone

$$\chi(C) = \max_q r_q(C),$$

$$r_q(C) = \begin{cases} B_q(C)/\varepsilon_q, & \varepsilon_q > 0, \\ 0, & \varepsilon_q = B_q(C) = 0, \\ +\infty, & \varepsilon_q = 0 < B_q(C). \end{cases} \quad (10)$$

The decision boundary is $\chi(C) = 1$. Values at or below one are certificate-feasible; a value above one means that at least one protected output is supercritical and the cone must grow or fall back to FULL. We use the phrase *criticality-bounded* in this narrow sense of a normalized output-error boundary. It is not meant to suggest a new model of physical critical phenomena.

4.5 Gaussian display-space bound

For a projected Gaussian i at pixel p , MAVEB starts from the projected opacity model

$$\alpha_i(p) = o_i \exp\left(-\frac{1}{2} q_i(p)\right), \quad (11)$$

where o_i is peak opacity and $q_i(p)$ is the squared Mahalanobis distance in projected Gaussian space. In the bound that follows, α_i means the effective value after the renderer's clamp

and cutoff rules, not simply the raw value from Eq. 11. For an edited set E , we define the aggregate opacity mass

$$A_p(E) = 1 - \prod_{i \in E} (1 - \alpha_i(p)). \quad (12)$$

Proposition 1 (Gaussian finite-edit display bound). Let U denote the unchanged Gaussian state and E_0, E_1 the complete before/after edited sets at a protected pixel p . Under the assumptions below,

$$\begin{aligned} \Delta_p &= \|R_p(U \cup E_0) - R_p(U \cup E_1)\|_\infty, \\ \Delta_p &\leq C_{\text{color}} \min(1, A_p(E_0) + A_p(E_1)). \end{aligned} \quad (13)$$

Assumptions. The proposition rests on four conditions. First, compositing is front-to-back, with effective post-clamp/post-cutoff alpha in $[0, 1]$. Second, each protected color channel, including the background and the renderer’s bounded spherical-harmonic evaluation, lies in an interval of width C_{color} . Third, E_0 and E_1 include every Gaussian whose projected opacity, color, insertion/deletion state, or depth relation to unchanged support can change because of the edit. Finally, renderer behavior that is not captured by these bounded quantities is treated as unsupported and is promoted to an exact dependency or a FULL fallback.

Proof sketch. For a single edited set, front-to-back compositing can alter a channel by no more than the channel range times the opacity mass through which edited support replaces, attenuates, or reveals unchanged radiance. This gives $C_{\text{color}} A_p(E)$. Insertion and deletion are the special cases in which one edited set is empty. For a general before/after edit, we pass through the common unchanged state U and apply the triangle inequality, obtaining $C_{\text{color}}(A_p(E_0) + A_p(E_1))$. The result is then capped by the largest possible channel-range change, C_{color} . The image-level certificate takes the maximum of Eq. 13 over the declared protected pixels. The supplement gives the algebra in more detail.

Opacity-only delta refinement. Equation 13 is intentionally general, but it is insensitive to how small an opacity-only edit is: the before/after opacity masses can remain large even when their difference is tiny. For paired Gaussian records whose position, projected covariance, depth order, rotation, and spherical-harmonic color are unchanged, the implementation uses the compositor’s coordinate-wise Lipschitz property in effective alpha. Let $\alpha_i^0(p)$ and $\alpha_i^1(p)$ be the renderer-admitted before/after alpha values. On pixels where at least one admitted alpha changes,

$$\Delta_p \leq C_{\text{color}} \min\left(1, \sum_i |\alpha_i^0(p) - \alpha_i^1(p)| + 0.002\right). \quad (14)$$

The 0.002 term covers the reference compositor’s two-sided early-termination discrepancy: each endpoint may omit at most 0.001 transmittance after accumulated alpha exceeds 0.999. The allowance is added only where an opacity delta is actually admitted, so pixels outside changed support retain a zero bound. The routine fails closed if any paired primitive changes geometry, rotation, covariance, spherical-harmonic

coefficients, or camera visibility unexpectedly. V5 and v6 use this refinement for opacity residuals; their translation control continues to use Eq. 13.

4.6 Stable temporal history

For stable temporal validation, let E_c denote current-frame error, E_h retained-history error, E_n neighborhood or clamping error, and $w \in [0, 1]$ the history weight. With $E_r = \max(E_h, E_n)$,

$$E_{\text{resolved}} \leq (1 - w)E_c + wE_r. \quad (15)$$

If validation, reprojection, or disocclusion can itself change, MAVEB does not apply Eq. 15; it promotes the relation to HARD invalidation. The temporal case is a useful example of the design rule we follow throughout: a soft bound is used only for as long as its own preconditions can be certified.

4.7 Greedy feasible-cone search and crossover

The v1 search still needs a numerical ordering when a declared tolerance is zero. For stable exteriors it uses $\tilde{\chi}(C) = \max_q B_q(C) / \max(\varepsilon_q, 10^{-15})$ only as a search score. The floor does not relax acceptance, which still uses $B_q(C) \leq \varepsilon_q$. For an expansion $C \rightarrow C'$ with positive added work, the utility is

$$\text{utility}(C \rightarrow C') = \frac{\tilde{\chi}(C) - \tilde{\chi}(C')}{W(C') - W(C)}. \quad (16)$$

Each candidate is closed over its exact predecessors before it is certified again. The search continues until it finds a cone that is both certificate-feasible and cheaper than FULL; if no such cone is reached, the planner returns the full fallback. This procedure gives us a certified feasible cone. We do not claim that it is the globally cheapest safe cone.

5 SYSTEM IMPLEMENTATION

Throughout this section, MAVEB names the experimental CBRC research system and AETHER names its underlying native persistent-world software stack. MAVEB keeps the scientific reference path separate from the production-oriented AETHER execution path. The dense Python certifier follows the mathematical contract directly and serves as a falsification and regression reference. The native C++23 AETHER implementation uses a sparse typed revision graph, sparse analytic propagation over supported DAG exteriors, exact predecessor checks, greedy feasible-cone search, and deterministic certificate serialization. For the frozen campaign, the bound production output-cone graph is evaluated by both the native planner and the Python reference planner; their final decisions are compared rather than assumed equivalent.

Revision transactions. The production path records persistent Gaussian entity translations, immutable before/after sidecars, exact camera binding, a production image certificate, and the planner’s temporal repair or retention decision. In headless operation, the system loads a persistent world, applies indexed Gaussian transactions, writes the new durable revision and sidecars, evaluates the Gaussian certificate, runs

the output-cone planner, and emits canonical evidence JSON. If the consistency checks do not pass, the revision fails closed instead of publishing a partly updated world.

Cross-derived-state accounting. A **LocalityLedger** tracks observation, TSDF, mesh, texture-page, material, Gaussian, GPU-publication, and temporal work; ledger/mesher disagreement is an error. Counts retain their native units until a versioned calibration is applied. The frozen public campaign calibrates only four measured domains: Gaussian inspection, Gaussian update, GPU publication bytes, and temporal invalidation pixels.

Independent oracle. The Gaussian oracle is independent of the planner. It renders before and full-after states, then forms a support replay from full-after values on the selected support and before-state values outside it. The per-pixel exterior residual is checked against Eq. 1; certificate disagreement or a residual above the emitted bound is a correctness failure. Evidence bundles record hashes, workflow IDs, and native/reference output-cone parity.

Representation scope. The paper-grade campaign contains two representation classes. Seventy-two scenes from 3RScan [24], ARKitScenes [5], and the Bonn RGB-D Dynamic Dataset [21] use dataset-native geometry seeded into deterministic Gaussian fields after the public capture assets are normalized into the benchmark coordinate convention. Thirteen scenes use pinned pretrained GraphDECO 3DGS representations from the standard evaluation suite: nine Mip-NeRF 360 scenes [4], Train and Truck from Tanks & Temples [15], and Dr Johnson and Playroom from Deep Blending [11]; the Gaussian representations preserve spherical-harmonic coefficients, opacity, anisotropic scale, and rotation [14]. Ownership is deterministic and spatial in both paths; it represents revision bookkeeping rather than semantic segmentation. Selected-to-FULL render fidelity is measured within the evaluated representation, separately from reconstruction quality against original camera photographs.

6 EVALUATION

6.1 Questions and frozen protocol

The paper-grade evaluation asks six questions. RQ1 tests whether every published LOCAL decision satisfies the emitted certificate under independent full-after replay. RQ2 measures how often the fail-closed rule selects LOCAL versus FULL as coupling increases. RQ3 measures calibrated selected work relative to the frozen full baseline. RQ4 tests whether the behavior persists across datasets and across seeded versus trained Gaussian representations. RQ5 tests translation, rotation, uniform-scale, and opacity edits under the same revision contract. RQ6 compares the resulting decisions with full, exact, fixed-radius, changed-fraction, empirical-order, and mechanism-ablation policies.

The campaign is frozen before outcome inspection. It contains 1,275 revisions over 85 scenes: 600 cases on 40 3RScan reference/rescan worlds [24], 300 cases on 20 ARKitScenes

worlds [5], 180 cases on 12 Bonn dynamic RGB-D sequences [21], and 195 cases on 13 pinned pretrained GraphDECO 3DGS scenes drawn from Mip-NeRF 360, Tanks & Temples, and Deep Blending [4, 11, 14, 15]. Every scene contributes the same 15-case coupling template, while edit family rotates through translation, rotation, uniform scale, and opacity. The frozen protocol therefore separates outcome inspection from threshold or case construction.

Each case records the source edit, chosen cone, decision (LOCAL or FULL), certificate, independent full-after replay, work, baselines, ablations, and provenance. Fallbacks stay in the dataset because they are part of the method’s behavior. The final visual audit compares the selected output with the independent FULL-after render at the same camera.

Work metric. The headline work ratio uses the frozen heterogeneous work model from Eq. 4. The four calibrated domains are Gaussian inspection, Gaussian update, GPU publication bytes, and temporal invalidation pixels. Bootstrap intervals use 5,000 resamples. The work ratio is an execution-independent accounting metric; paired local-versus-full wall-clock timing remains outside the current claim.

Statistical reporting. Local and violation rates in the broad campaign use Wilson 95% intervals. Median work ratios use bootstrap 95% intervals. Paired baseline comparisons count cases in which both policies certify, then report which policy uses less calibrated work, ties, and an exact two-sided sign test when the comparison is informative. The confirmatory residual study uses the scene as its primary sampling unit. Its 95% intervals resample scenes within each fixed dataset stratum, preserving the five-scene-per-dataset design in every bootstrap replicate; the 3,840 parameter cases are pooled evidence, not 3,840 independent samples.

6.2 Residual-sensitive protocol

The non-zero-residual experiments are frozen separately from the 1,275-case base campaign. The v4 diagnostic uses four scenes, four locality profiles, exact and 1/1024 residuals, and an epsilon ladder of 1, 4, 16, 32 divided by 255. It is retained unchanged after inspection. V5 changes the image certificate only for opacity residuals: corresponding Gaussian records must be identical except for opacity logit, and the bound uses the admitted change in pixel alpha rather than the older magnitude-insensitive opacity envelope. Translation remains on the older envelope as a control. V5 uses four scenes, residuals exact, 1/4096, 1/1024, and 1/256, and six epsilon values from 1/255 to 32/255.

V6 is a confirmatory breadth run, not a new method. It repeats the v5 mechanism without changing thresholds, residual scales, edit magnitudes, locality profiles, or control family. Five deterministically ranked scenes are selected from each of 3RScan, ARKitScenes, Bonn RGB-D Dynamic, and trained GraphDECO 3DGS, giving 20 scenes and 3,840 pooled parameter cases. Confirmation gates are fixed before execution.

Failed scenes or cases are not replaced after outcome inspection.

6.3 Baselines and ablations

The method-level suite includes FULL, exact-only repair, fixed graph radii, changed-fraction selection, empirical ordering followed by analytic re-certification, and CBRC. Structural ablations remove individual mechanisms while retaining the frozen campaign. We preserve neutral ablations as neutral results. A mechanism can remain important to the contract even when this real matrix does not separate its aggregate work from the full method; deterministic mechanism-isolation tests are retained separately for falsifiability.

7 RESULTS

7.1 Overall paper-grade campaign

All 1,275 frozen cases complete across all 85 scenes. CBRC selects 935 certified LOCAL repairs and 340 FULL fallbacks, giving a local rate of 73.33% (95% CI 70.84–75.69%). We observe zero certificate violations; the corresponding 95% Wilson interval for the violation rate is 0–0.3004%. Both local and fallback outcomes appear in every representation class.

The median calibrated selected work ratio is

$$\text{median } W(C)/W_{\text{full}} = 0.4744733, \quad (17)$$

with bootstrap 95% CI [0.46657, 0.47885]. At the median, this is 52.55% lower calibrated work than FULL. The minimum observed ratio is 0.07030; fallback cases have ratio 1 by construction.

The identical 73.33% local rate across datasets is a consequence of applying the same frozen 15-case coupling template to every scene; it should not be interpreted as evidence that the datasets behave identically. Dataset-dependent work differs substantially: the median ratio is 0.341 on 3RScan and 0.519 on trained 3DGS.

7.2 Coupling controls fallback behavior

The fail-closed envelope becomes visible when the campaign is grouped by coupling regime. All low and medium cases remain local. High coupling produces 170 local repairs and 85 fallbacks, while adversarial coupling produces 85 local repairs and 255 fallbacks. The pattern is therefore monotonic in this frozen matrix rather than an all-success story.

7.3 Generalization across edit families

The same revision contract is exercised across four edit families. Opacity remains local in all 255 cases. Translation and rotation each remain local in 255 of 340 cases. Uniform scale is the hardest family in the frozen matrix, with 170 local repairs and 170 fallbacks. No edit family contains an observed certificate violation.

7.4 Independent selected-render audit

The selected render is byte-identical to the independent FULL-after render in all 1,275 cases on the evaluated benchmark views: the selected exact-case rate is 1.0. For fallback cases this equality follows from selecting FULL. For the 935 LOCAL cases it means the executed repair reproduces the full-after raster exactly at 8-bit RGB precision on the audited view.

This result is strong evidence for the selection/fallback implementation, but it is deliberately separated from certificate tightness. Because the measured selected-to-FULL residual is zero throughout this broad campaign, the matrix does not demonstrate a useful case with

$$0 < E_q^{\text{full-ref}} \leq B_q(C) \leq \varepsilon_q. \quad (18)$$

The separately frozen residual-sensitive sequence below targets exactly this regime without changing the broad campaign.

7.5 From diagnostic failure to confirmatory non-zero locality

The first residual-sensitive freeze, v4, is a negative result. Across 512 cases, the 256 exact-residual cases remain local, while every one of the 256 non-zero-residual cases falls back to FULL. There are no certified non-zero local cases and no tolerance crossovers. Postmortem analysis attributes all 256 non-zero fallbacks to the post-repair certified bound exceeding epsilon. The outcome is insensitive to the locality profile and to the tested epsilon ladder. This rules out a convenient interpretation in which the test scenes were simply too globally coupled: the bottleneck is the certificate used for the residual.

V5 changes one mechanism. For opacity-only residuals, geometry, covariance, rotation, depth, and spherical-harmonic color remain fixed, while the certificate accumulates the actual change in admitted per-pixel alpha with a conservative transmittance allowance. Translation continues to use the old opacity-envelope certificate. All certificate-routing, monotonicity, selected-output, and repair-certificate validity gates pass. Of 384 opacity cases, 251 select LOCAL and 133 select FULL; 142 are certified non-zero local cases and 29 stress keys cross from full to local as epsilon changes. The 384 translation controls contain 96 exact local cases and 288 full fallbacks, with zero non-zero local cases and zero crossovers. The contrast identifies certificate sensitivity rather than a generic tendency of the stress matrix to produce locality.

V6 repeats the same mechanism across 20 independent scenes. The confirmatory audit passes every predeclared gate, with no selected-certificate or repair-certificate violation. Non-zero opacity locality occurs in all 20 scenes and all four dataset families; opacity crossover also occurs in all 20 scenes. Across scenes, the mean opacity non-zero-local rate is 0.477, with a dataset-stratified scene-bootstrap 95% interval of [0.432, 0.517]. Translation remains at 0.000 [0.000, 0.000], so the paired opacity-minus-translation difference is

Table 2. Paper-grade results by dataset/representation. Work is calibrated selected work divided by the frozen FULL baseline. Confidence intervals are Wilson intervals for local rate and bootstrap intervals for median work.

Dataset / representation	Cases	LOCAL	FULL	Local rate (95% CI)	Median work/FULL (95% CI)
3RScan	600	440	160	0.733 [0.697, 0.767]	0.341 [0.337, 0.341]
ARKitScenes	300	220	80	0.733 [0.681, 0.780]	0.483 [0.480, 0.494]
Bonn RGB-D Dynamic	180	132	48	0.733 [0.664, 0.793]	0.475 [0.474, 0.488]
Trained GraphDECO 3DGS	195	143	52	0.733 [0.667, 0.790]	0.519 [0.513, 0.541]
Overall	1275	935	340	0.733 [0.708, 0.757]	0.474 [0.467, 0.479]

Table 3. Decision envelope by coupling regime.

Regime	Cases	LOCAL	FULL	Median work/FULL
Low	340	340	0	0.415
Medium	340	340	0	0.445
High	255	170	85	0.510
Adversarial	340	85	255	1.000

Table 4. Paper-grade outcomes by edit family.

Edit	Cases	LOCAL	FULL	Median work/FULL
Opacity	255	255	0	0.457
Rotation	340	255	85	0.453
Translation	340	255	85	0.475
Uniform scale	340	170	170	0.770

numerically the same 0.477 [0.432, 0.517]. The generic audit contains 687 certified non-zero local cases and 151 tolerance-crossover groups.

The same-opacity legacy-envelope counterfactual provides a direct mechanism check in v6. Holding the physical opacity edit fixed, the delta-sensitive certificate rescues a mean 0.531 of cases per scene that the legacy envelope would reject (95% interval [0.512, 0.549]); all 20 scenes contain at least one such rescue. The median scene-level ratio of legacy residual bound to delta-sensitive bound is 122.3. This comparison does not claim that the new certificate is tight. The accepted cases remain far from the tolerance boundary: the largest scene maximum of measured residual divided by epsilon is 0.0812, the median of those scene maxima is 0.0195, and there are no near-boundary local cases. The median scene-level residual effectivity is approximately 1168, so conservatism remains a measured limitation rather than being hidden by the positive locality result.

7.6 Baselines and ablations

The paired statistics expose both useful separation and neutral mechanisms. Against FULL, all 1,275 cases are jointly certified; CBRC uses lower calibrated work in 935 and ties in 340, with exact two-sided sign-test $p = 6.89 \times 10^{-282}$. Replacing the exterior tail with a global-norm bound produces the same 935-versus-340 separation, showing that the localized tail is the dominant source of work reduction on this matrix.

Several comparisons are neutral. Empirical ordering followed by analytic re-certification ties CBRC in all 1,275

jointly certified cases. Five ablations also tie on aggregate work: Gaussian bound, hard/soft distinction, predecessor closure, QoI specialization, and temporal bound. Exact-only and fixed-radius policies certify 340 of 1,275 cases. The changed-fraction policy certifies 537. Removing fallback leaves 340 certified cases. We retain these results because the experiment characterizes the contract instead of forcing every mechanism to separate.

8 DISCUSSION

8.1 What the evidence establishes

The broad campaign establishes the normal operating envelope: the fail-closed rule survives independent replay without an observed certificate violation in 1,275 cases, produces both local repair and full fallback across four datasets and two representation classes, and reduces calibrated selected work to a median 0.474 of FULL. The residual-sensitive sequence establishes a different point. V4 shows that a magnitude-insensitive image certificate can erase useful locality even when the repair itself is graded. V5 isolates a certificate change that restores non-zero locality for opacity while the translation control remains unchanged. V6 then repeats that mechanism across 20 scenes and passes the predeclared scene-level confirmation gates. The positive v6 result therefore depends on a specific, testable change in the certificate rather than on post-hoc scene or tolerance selection.

The result is best understood as a systems contract rather than as a new dependency algorithm in isolation. Dependency-aware recomputation, self-adjusting computation, norm bounds, and goal-oriented error estimation provide important prior ideas. CBRC specializes them into a persistent captured-world rule that requires exact structural closure, conservative exterior accounting, output-specific tolerances, a frozen work crossover, explicit fallback, immutable evidence, and independent full-after falsification.

8.2 Limitations

Correctness and certificate tightness remain separate. The broad campaign is exact at the audited 8-bit raster, while v5 and v6 demonstrate certified non-zero residuals. That closes the earlier evidence gap but does not make the certificate tight. In v6, no accepted case lies near the tolerance boundary; the maximum scene-level measured-residual-to-epsilon ratio is 0.0812, and the median scene-level residual

Table 5. Residual-sensitive evidence sequence. Pooled case counts describe frozen parameter grids; v6 inference uses 20 scenes as the primary sampling units, stratified five per dataset. Non-zero LOCAL means a selected local repair with a measured non-zero residual that remains within its certified bound and tolerance.

Stage	Frozen population	Non-zero LOCAL evidence	Main diagnostic / control result
v4 diagnostic	4 scenes, 512 cases	0 cases; 0 tolerance crossovers	All 256 non-zero-residual cases fall back because the certified post-repair bound exceeds epsilon.
v5 mechanism	4 scenes, 768 cases	142 opacity cases across all 4 datasets; 29 crossover keys	Opacity: 251 LOCAL/133 FULL. Translation control: 0 non-zero LOCAL, 0 crossovers.
v6 confirmation	20 scenes, 3,840 cases	687 cases; all 20 scenes and all 4 datasets; 151 crossover groups	Scene-level opacity-translation difference 0.477, 95% CI [0.432, 0.517]; all frozen gates pass.

Table 6. Scene-level v6 confirmation. Intervals are dataset-stratified scene-bootstrap 95% intervals over 20 scenes; pooled case totals are not treated as independent samples.

Metric	Result
Scenes with non-zero opacity LOCAL	20 / 20
Scenes with opacity tolerance crossover	20 / 20
Opacity non-zero-local rate	0.477 [0.432, 0.517]
Translation non-zero-local rate	0.000 [0.000, 0.000]
Same-opacity legacy-envelope rescue rate	0.531 [0.512, 0.549]
Largest scene maximum of residual / epsilon	0.0812
Median scene residual effectivity	1168.2

effectivity is approximately 1168. These values show substantial safety margin and also substantial conservatism. We therefore claim repeatable residual-sensitive locality under the tested opacity-only assumptions, not a generally tight display-space error estimator.

Calibrated work is distinct from runtime. The 0.474 median headline is a frozen four-domain work ratio, not a paired end-to-end speedup. It captures heterogeneous work under a versioned calibration but omits effects such as allocator behavior, synchronization, cache locality, command submission, and pipeline overlap. A runtime claim requires matched LOCAL and FULL execution under the same conditions.

Representation coverage remains output-side. The broad campaign is materially wider than the earlier four-scene experiment: it contains 72 dataset-native geometry worlds and 13 trained 3DGS scenes. Even so, the measured revision path concentrates on Gaussian inspection/update, GPU publication, and temporal output invalidation. The wider observation/TSDF/mesh/texture/material ledger is implemented and regression-tested but is not yet represented by an equally broad paper-grade performance campaign.

Dataset-normalized worlds are not a reconstruction benchmark. For 3RScan, ARKitScenes, and Bonn, the benchmark evaluates deterministic seeded Gaussian representations derived from public captured-world assets. The trained GraphDECO subset evaluates pinned trained 3DGS. Selected-to-FULL equality is therefore a revision-fidelity result within each evaluated representation, rather than a claim

about photorealistic reconstruction quality against held-out photographs.

Some mechanisms are weakly identifiable on the real matrix. Several ablations tie the full method on aggregate work and certification. We keep those ties visible. The mechanism-isolation tests establish that those components can become consequential under controlled stress, while the paper-grade matrix shows that they do not dominate the measured real-scene outcomes.

The planner is deliberately conservative. Uniform-scale and adversarial cases trigger many full rebuilds. Those fallbacks are part of the intended behavior: when exterior influence, temporal stability, or work crossover cannot support a local result, the system prefers a full revision. The paper does not infer global optimality from the selected cones.

8.3 Implications

The main practical lesson is that locality should be treated as a contract, not as a geometric assumption. A small physical edit is a useful starting cue, but the publishable decision depends on the dependency closure, the output quantity being protected, the tolerated residual, and the cost of carrying the repair through derived state. This framing also makes failure informative. A FULL result identifies a revision for which the current model cannot justify leaving enough of the world untouched.

The residual-sensitive experiments make this distinction concrete. A local result can differ measurably from the full-after result and still be admissible when the measured residual remains below both the conservative bound and the declared tolerance. The v4 failure is as informative as the later success: locality is not created merely by allowing error; the certificate must represent the magnitude of the residual. V5 and v6 show that this distinction is repeatable for opacity-only residuals across the tested datasets. The remaining problem is not whether non-zero locality can occur, but how much the conservative bound can be tightened without losing the fail-closed guarantee.

9 REPRODUCIBILITY AND ARTIFACT SCOPE

The base completion manifest binds the 1,275 campaign rows, 85 prepared worlds, dataset import manifests, frozen case

matrix, work calibration, visual audit, cross-dataset statistics, tool fingerprints, and evidence Git SHA. The residual-sensitive line keeps v4 and v5 immutable and freezes v6 independently at 20 scenes and 3,840 cases. Its retained evidence includes the frozen campaign and provenance, canonical rows and baselines, scene-level metrics, the generic and v6 audits, the manuscript claim packet, case-decision trace, and deterministic reviewer-figure index. Cases resume only under matching deterministic fingerprints; bulky per-case render material is reproducible and may be compacted after the canonical row, baseline, replay manifest, and completion marker are retained.

Public datasets remain upstream assets rather than being redistributed with the repository. The repository contains the dense reference certifier, native planner and Gaussian oracle, trained and seeded representation paths, baseline and ablation suites, deterministic statistics and visual-audit generators, residual-sensitive certificate tests, regression tests, and storage-safe campaign runners. For double-blind review, identifiable repository and author metadata should be replaced with an anonymized frozen supplement that follows the target venue’s policy.

10 CONCLUSION

Criticality-Bounded Revision Cones give persistent 3D systems a way to avoid unnecessary global rebuilds without treating spatial locality as a safe assumption. A repair proceeds only when structural dependencies are closed, the effect left outside the selected region stays within the declared output tolerance, and the work remains worthwhile; otherwise MAVEB rebuilds. Across 1,275 revisions over 85 scenes, this policy produces both selective maintenance and deliberate fallback with zero observed certificate violations, while median calibrated work is 0.474 of FULL. Stronger coupling reduces the opportunity for reuse rather than forcing an unsafe local answer.

The residual-sensitive sequence resolves the main limitation of that exact-output campaign. The immutable v4 diagnostic showed that the original image certificate rejected every non-zero residual. A delta-sensitive opacity certificate restored certified non-zero locality in v5, and v6 repeated the unchanged mechanism across 20 scenes: all 20 showed non-zero opacity locality and tolerance crossover, the scene-level opacity-minus-translation difference was 0.477 with 95% interval [0.432, 0.517], and every confirmatory validity gate passed without an observed certificate violation. The same evidence also shows what remains unfinished: the certificate is conservative and the accepted cases are far from the tolerance boundary. The practical claim is therefore bounded and testable. Recomputation can be reduced when the current dependency and error model can justify it, and expanded when it cannot.

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Generative AI use disclosure. OpenAI ChatGPT was used for literature-search support, software and scripting assistance, debugging, experiment planning, visualization and manuscript tooling, and drafting or language editing. Reported measurements and frozen evidence came from project software and experimental artifacts. The author is responsible for the code, citations, analysis, claims, conclusions, and final text; no generative AI system is listed as an author.

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